A Theoretical Framework for Representing Ideological Bias through Sentiment Analysis

Sebastián Pinto (spinto@df.uba.ar), Lucas Díaz Celauro, Juana Rodríguez Gallo, Pablo Balenzuela, Matteo Serafino y Hernán A. Makse.

Universidad de Buenos Aires & CONICET — Instituto de Física Interdisciplinaria y Aplicada (IFIBA), Buenos Aires, Argentina · Independent Researcher, Corsano, Italy · Levich Institute & Physics Dept., City College of New York, USA.

ABSTRACT

Media bias, defined as the systematic tendency of outlets to favor or oppose particular political actors, shapes how audiences interpret political reality. Existing approaches infer ideology from textual similarity or from patterns of interaction and diffusion online, but often rely on indirect proxies that limit interpretability. In this work, we introduce a **maximum-entropy framework** to infer the ideological positions of media outlets and political figures from the distribution of **positive, neutral, and negative mentions** observed in online communication. Applied to Facebook posts from the **2024 U.S.** and **2023 Argentine** presidential elections, the model cleanly separates candidates along ideological dimensions and recovers consistent groupings of media outlets without any prior knowledge of ideological orientation, relying solely on observed sentiment, and validated against independent external classifications of media bias.

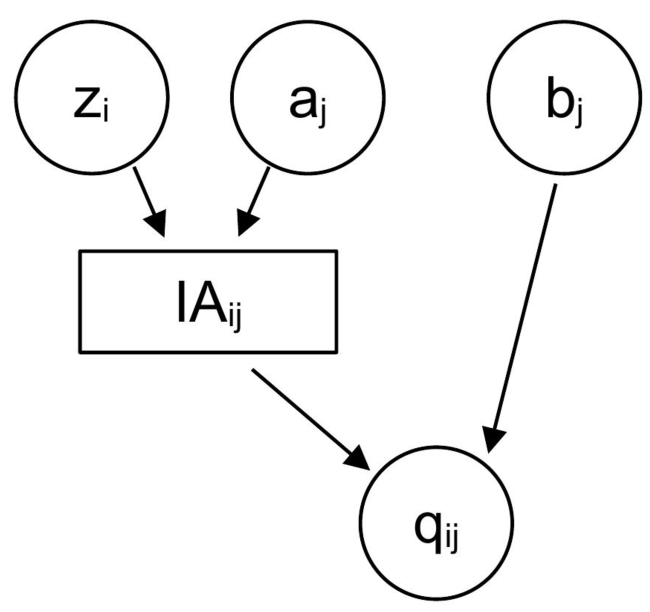
MODEL

Each mention of subject j by outlet i is positive, neutral, or negative (c = +1, 0, −1). Maximizing entropy subject to a constraint on the expected sentiment yields a **Boltzmann-like distribution**:

$$P(c\,|\,q) = \frac{e^{cq}}{Q_1}, \quad Q_1 = e^{q} + 1 + e^{-q}$$

For m outlets and k subjects, each outlet has an ideological position $z_i \in \mathbb{R}^D$ and each subject $a_j \in \mathbb{R}^D$. The **ideological alignment** $IA_{ij} = a_j \cdot z_i$ combines with a subject-specific **context bias** b_j (e.g. a scandal or a policy win) into a single parameter:

$$q_{ij} = \mathbf{a}_j \cdot \mathbf{z}_i + b_j$$

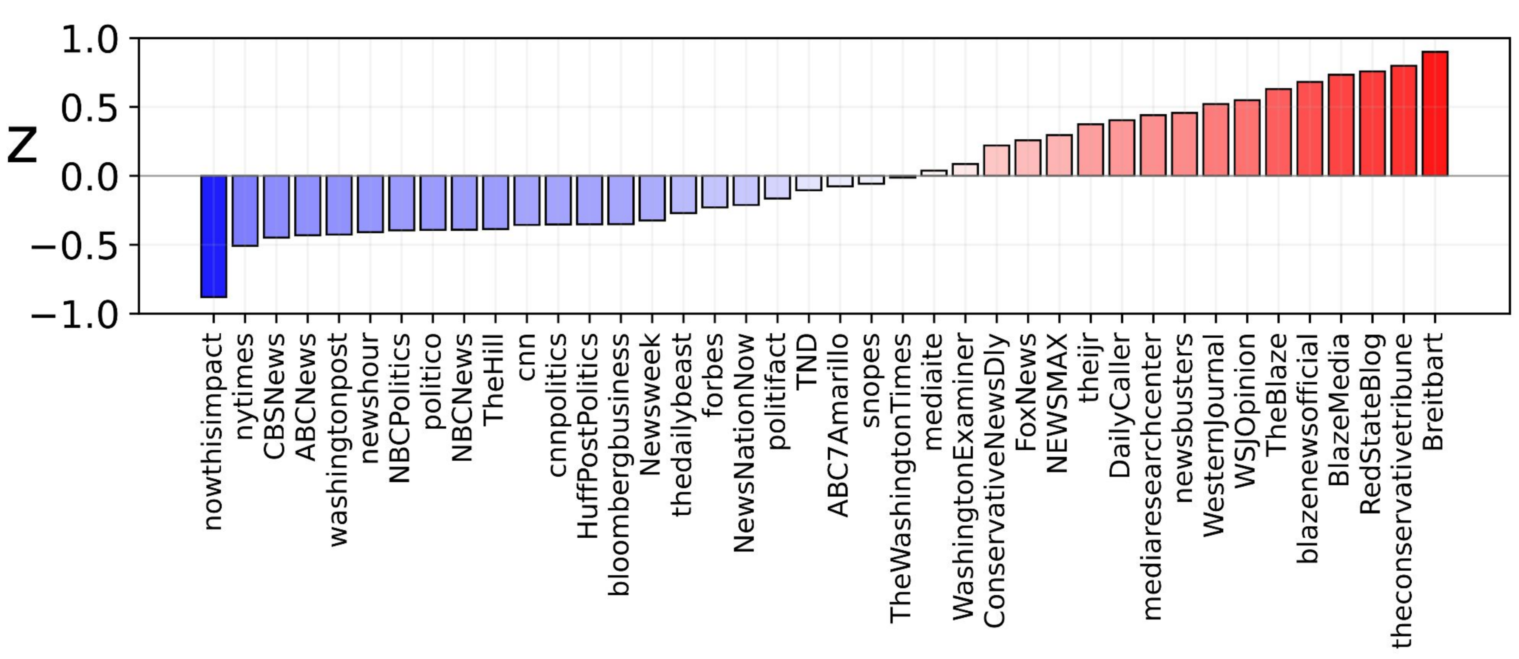


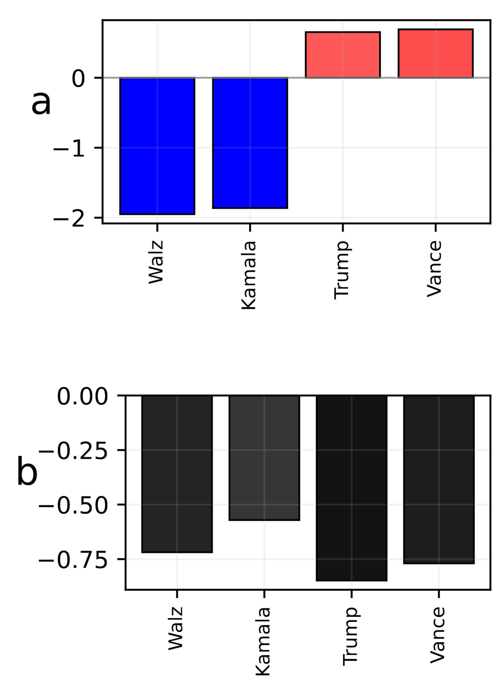
Positive q_{ij} ⇒ alignment / favorable context ⇒ more positive mentions. Negative q_{ij} ⇒ divergence / unfavorable context ⇒ mentions skew negative. Parameters (z, a, b) are fit by **maximum a posteriori** estimation over the full mention-count matrix N (Gaussian priors act as regularizers).

RESULTS — 2024 U.S. ELECTION

Dataset: ~38,000 Facebook posts (Aug–Nov 2024) from 40 leading U.S. media pages, each mentioning a single candidate: Kamala Harris, Tim Walz (Dem.) or Donald Trump, JD Vance (Rep.). A sentiment classifier labels each mention as positive, neutral, or negative.

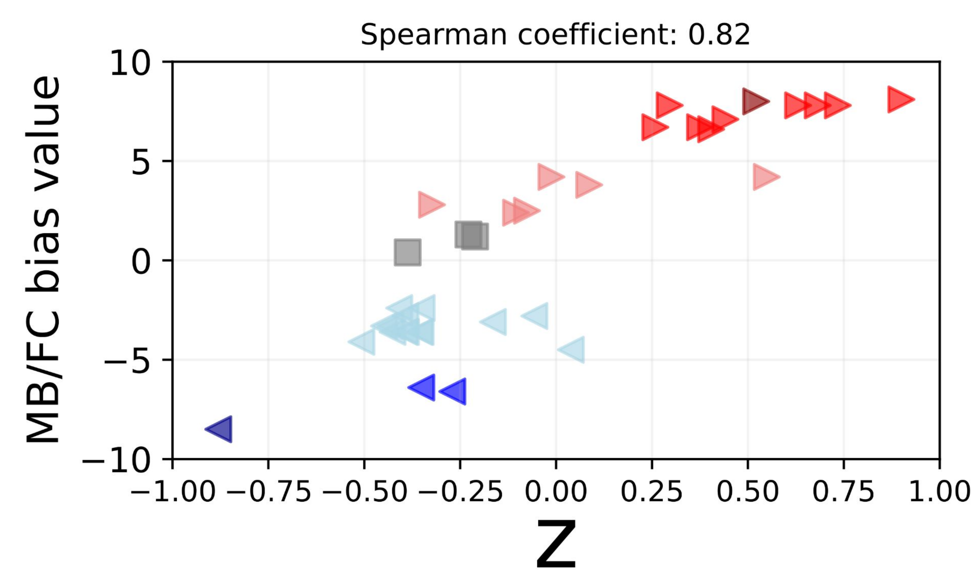
With two well-defined parties, a **1-dimensional model** (d = 1) already separates the political landscape: it cleanly distinguishes Democratic from Republican candidates and orders all 40 outlets along a left–right spectrum. The New York Times and Breitbart emerge at opposite extremes. All four candidates show **negative context bias** (b < 0): negative mentions dominate regardless of outlet.





VALIDATON OF 2024 U.S. ELECTION

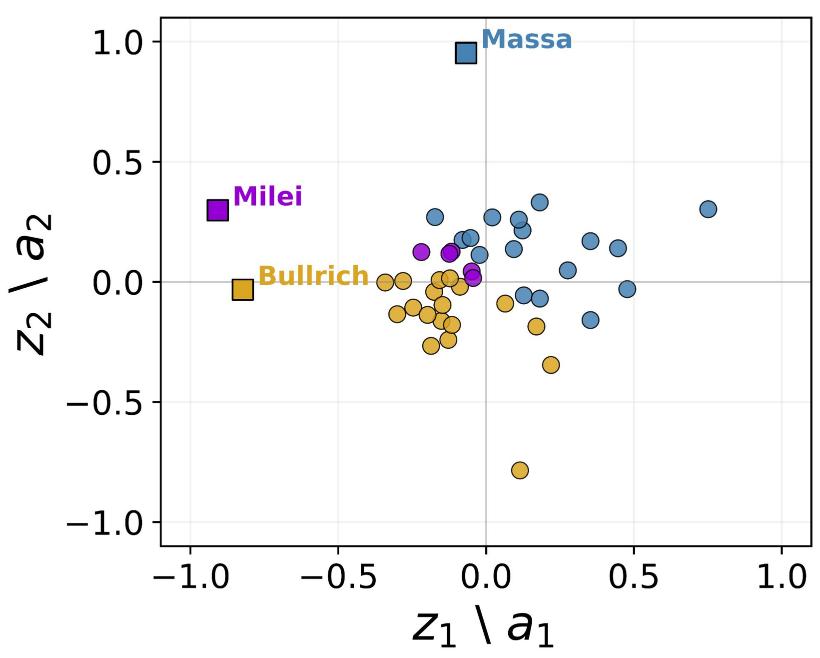
Inferred outlet positions z correlate strongly with independent **Media Bias / Fact Check** ratings (**Spearman ρ ≈ 0.82**), confirming the model recovers real ideological orientation from sentiment alone, with no external label ever used during inference.

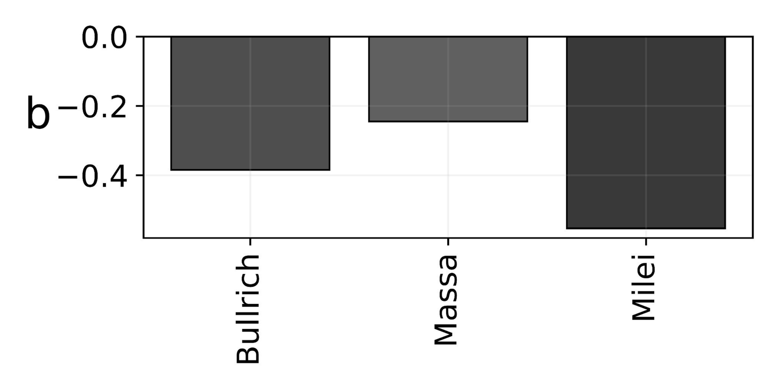


RESULTS — 2023 ARGENTINA ELECTION

Argentina's 2023 first round produced **three competitive candidates** (Massa 36.3%, Milei 30.2%, Bullrich 23.8%), requiring **d = 2** ideological dimensions: $q_{ij} = a_j \cdot z_i + b_j$, now a dot product of 2D vectors.

Using ~17,855 Spanish-language posts across 40 outlets (Aug–Oct 2023), the 2D fit shows most outlets clustering near the traditional candidates Massa and Bullrich, with only a few aligned to **Milei,** consistent with his profile as a political outsider openly critical of traditional media. All three candidates again show negative context bias, most pronounced for Milei.





CONCLUSIONS

This maximum-entropy framework infers interpretable, data-driven ideological positions from sentiment alone. No prior ideological labels are required. It scales naturally from a single left–right axis (U.S.) to **multi-dimensional** political landscapes (Argentina), and its outlet-level estimates align with independent bias classifications. Estimates remain stable even with **60% less data**.

Beyond classification, exposing latent media bias offers a practical tool for digital-literacy initiatives, helping audiences diversify consumption and counteract echo chambers. Future work will develop a fully probabilistic (Bayesian) version to quantify uncertainty in the inferred positions.

https://sophylab-df.github.io/